4 – Execute the Burgwald Data Pipeline

Clone · Configure · Run · Verify

Chris Reudenbach

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Goals

- Clone the course repository and open the RStudio project.
- Run all core pipeline scripts **unchanged** on your own machine.
- Verify that raw data, processed data, and predictor stacks are created.
- Document any problems that prevent full reproducibility.

Repository

Clone this repository locally and open the `.Rproj`:

https://github.com/gisma/LV-19-d19-006-25-R.git

Task 1 – Basic project layout

After cloning and opening the project, check that these top-level folders exist (create if missing):

```
data-pipeline
   Setup
         01_setup-burgwald.R
   Generic data retrievale
         01-data-retrieval.R (plus helpers)
   External terrain tools
         01-04-1-SAGA.Rexecute:
  cache: false
   Sentinel-2 (CDSE)
         01-2-CDSE-sentinel-data-retrieval.R
   Sentinel-2 (gdalcubes)
         01-3-gdalcubes-sentinel-data-retrieval.R
   Output
          NetCDF predictor stacks in data/predictor/
```

Task 2 – Run the setup

Script: `01_setup-burgwald.R`

- Open the script.
- Run it from top to bottom **without any modifications**.
- After successful execution, the following must exist:

```
data/raw/AOI_Burgwald/
data/processed/aoi_burgwald.gpkg
src/01-fun-data-retrieval.R  (loaded without error)
```

Minimal check (optional in console):

```r
fs::dir_ls("data", recurse = TRUE)
```

Task 3 – Run generic data retrieval

Script: `01-data-retrieval.R`

- Run the script unchanged.
- Afterwards, the following data must physically exist (content not relevant, existence is):
  - DEM (GeoTIFF)
  - OSM-by-key files
  - CLC raster / GeoPackage
  - DWD station data in `data/raw/dwd-stations/`
  - processed DWD files in `data/processed/dwd/`

Suggested quick check:

```r
fs::dir_ls("data", recurse = TRUE)
```

Task 4 – Test external API (SAGA)

Script: `01-04-1-SAGA.R`

- Adjust the SAGA binary path if needed (only this, nothing else).
- Run the script.
- At least **one new terrain raster** (e.g. slope, aspect) has to be created.
- Note the file name and location.

Task 5 – Sentinel-2 via CDSE

Script: `01-2-CDSE-sentinel-data-retrieval.R`

- Run the script unchanged.
- Expected result (folder may vary slightly, adapt to repo):

```
data/raw/s2/
     <Sentinel-2 scenes or COG references>
```

- Count how many items were downloaded or referenced.
- Check if a STAC JSON file was created (if implemented in the script).

Task 6 – Sentinel-2 via gdalcubes

Script: `01-3-gdalcubes-sentinel-data-retrieval.R`

- Run the script unchanged.
- Expected output:

```
data/predictor/
      <NetCDF file>   # e.g. s2_predictor_stack_summer.nc
```

- Confirm that the NetCDF file exists and can be opened, e.g.:

```
stars::read_stars("data/predictor/<your-file>.nc")
```

Task 7 – Pipeline check

Run this final check:

```
list.files("data/raw", recurse = TRUE)
list.files("data/processed", recurse = TRUE)
list.files("data/predictor", recurse = TRUE)
```

Verify that all three stages exist on your system:

- RAW DATA
- PROCESSED DATA
- PREDICTOR STACK(S)

If any stage is missing, re-run the corresponding script or document the error.

Task 8 – Short reproducibility log

Write **one short paragraph** (max. 5–6 sentences) for your own log:

- Did the full pipeline run on your machine?
- Which scripts worked without issues?
- Where did you have to change paths or credentials (e.g. SAGA, CDSE)?
- Which errors (if any) stopped the pipeline?

You will use this later to compare reproducibility between systems.

Expected final structure

The expected directory tree at the end of this worksheet is:

```
data/
  raw/
    AOI_Burgwald/
    dem/
    osm_by_key/
    dwd-stations/
    s2/
  processed/
    aoi_burgwald.gpkg
    dwd/
  predictor/
    *.nc
src/
  setup/
  retrieval/
  remote_sensing/
outputs/
  figures/
  tables/
```

Take-home summary

This pipeline forms the foundation for all later modelling tasks in the course.By running it successfully, you ensure that:

all Burgwald base layers (DEM, OSM, CLC, DWD) are available,

all Sentinel-2 sources (CDSE + gdalcubes) are standardised and reproducible,

the project contains a clean, FAIR-aligned data structure, and

downstream scripts (classification, predictors, modelling) can run without modification.

In short: if this pipeline runs, every later analysis in the course becomes plug-and-play.